

CACS101: Computer Fundamentals and Applications

Unit 4 – Database Management System

PRAJWAL BISTA

Database

- A database is organized collection of related data of an organization stored in formatted way which is shared by multiple users.
- It must be well organized.
- It is related.
- It is accessible in a logical order without any difficulty.
- E.g. University database maintains information about students, courses and grades in university.

Introduction to DBMS

- The database, its processing methods and the set of rules and conditions to be followed; collectively known as database management system (DBMS).
- The primary goal of DBMS is to store and manage data both conveniently and efficiently.
- Database systems are generally designed to manage large volume of information.
- Management of data involves defining structure for storage of information and providing mechanisms for manipulation of information.

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- DBMS can also define as a general purpose software system that enables user to create, maintain and manipulate database.
- It provides fast and convenient access to information from data stored in database.
- DBMS interfaces with application programs so data contained in database can be accessed by multiple applications and users.
- Some popular DBMS softwares are: Oracle, SQL – Server, IBM-DB2, MySQL, MS Access, Sybase etc.

Applications of DBMS

- Banking: customer and their account info
- Airlines: reservations and schedules info
- Universities: student info, grades etc.
- Credit card transactions: for purchase on credit cards and generation of statements.
- Finance: for storing information about holding, sales and purchases etc.
- Manufacturing: for management of supply chain.
- Human resources: for information about employee

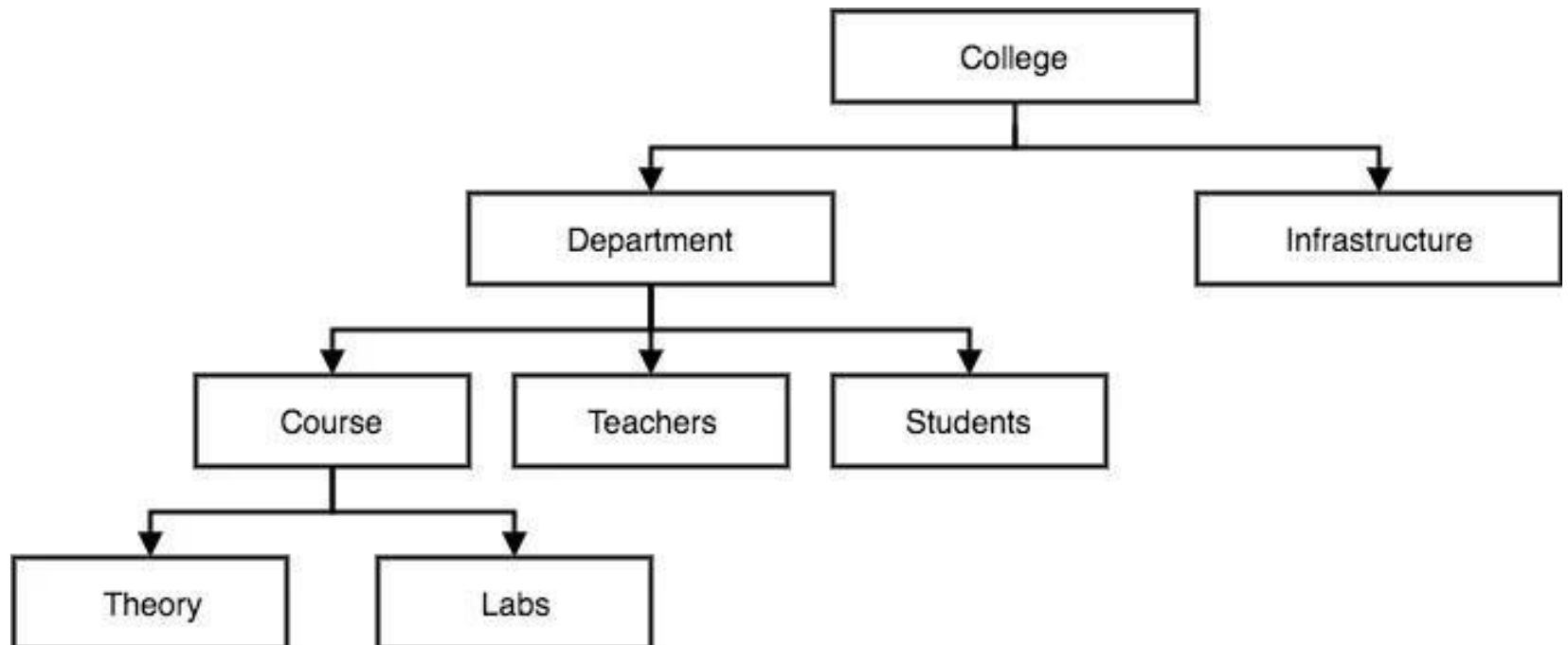
Database Models

- A Database model defines the logical design and structure of a database and defines how data will be stored, accessed and updated in a database management system.
 - Hierarchical Model
 - Network Model
 - Entity-relationship Model
 - Relational Mode
 - Object Oriented Data Model

Hierarchical Model

- This database model organises data into a tree-like-structure, with a single root, to which all the other data is linked.
- The hierarchy starts from the root data, and expands like a tree, adding child nodes to the parent nodes. In this model, a child node will only have a single parent node.
- In hierarchical model, data is organised into tree-like structure with one one-to-many relationship between two different types of data, for example, one department can have many courses, many professors and of-course many students.

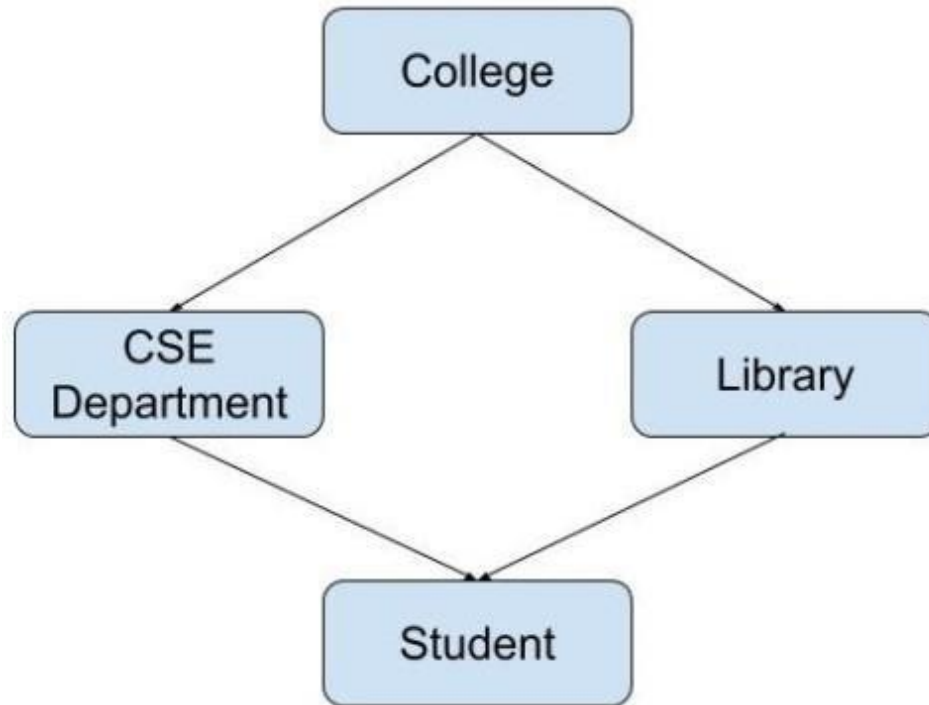
Hierarchical Model



Network Model

- This is an extension of the Hierarchical model. In this model data is organised more like a graph, and are allowed to have more than one parent node.
- In this database model data is more related as more relationships are established in this database model. Also, as the data is more related, hence accessing the data is also easier and fast.
- This database model was used to map many-to-many data relationships. This was the most widely used database model, before Relational Model was introduced.

Network Model



Network Model

Entity-relationship Model

- Entity-Relationship Model or simply ER Model is a high-level data model diagram.
- In this model, we represent the real-world problem in the pictorial form to make it easy for the stakeholders to understand. It is also very easy for the developers to understand the system by just looking at the ER diagram.
- We use the ER diagram as a visual tool to represent an ER Model.

Entity-relationship Model

- ER diagram has the following three components:
 - Entities: Entity is a real-world thing. It can be a person, place, or even a concept. Example: Teachers, Students, Course, Building, Department, etc are some of the entities of a School Management System.
 - Attributes: An entity contains a real-world property called attribute. This is the characteristics of that attribute. Example: The entity teacher has the property like teacher id, salary, age, etc.
 - Relationship: Relationship tells how two attributes are related. Example: Teacher works for a department.

Types of relationship

- One to One Relationship
- One to Many Relationship
- Many to One Relationship
- Many to Many Relationship

One to One Relationship

- When a single instance of an entity is associated with a single instance of another entity then it is called one to one relationship. For example, a person has only one passport and a passport is given to one person.



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One to Many Relationship

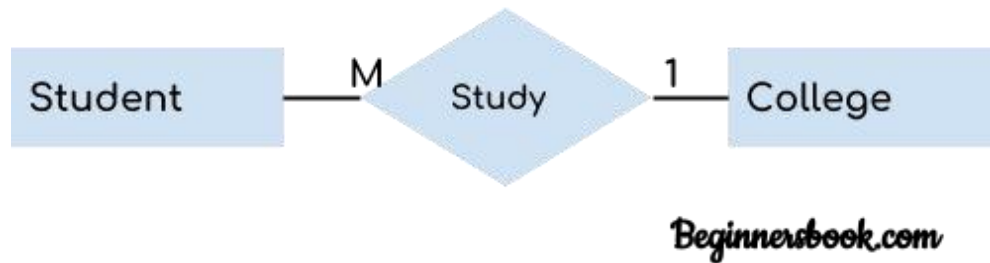
- When a single instance of an entity is associated with more than one instances of another entity then it is called one to many relationship. For example – a customer can place many orders but a order cannot be placed by many customers.



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Many to One Relationship

- When more than one instances of an entity is associated with a single instance of another entity then it is called many to one relationship. For example – many students can study in a single college but a student cannot study in many colleges at the same time.

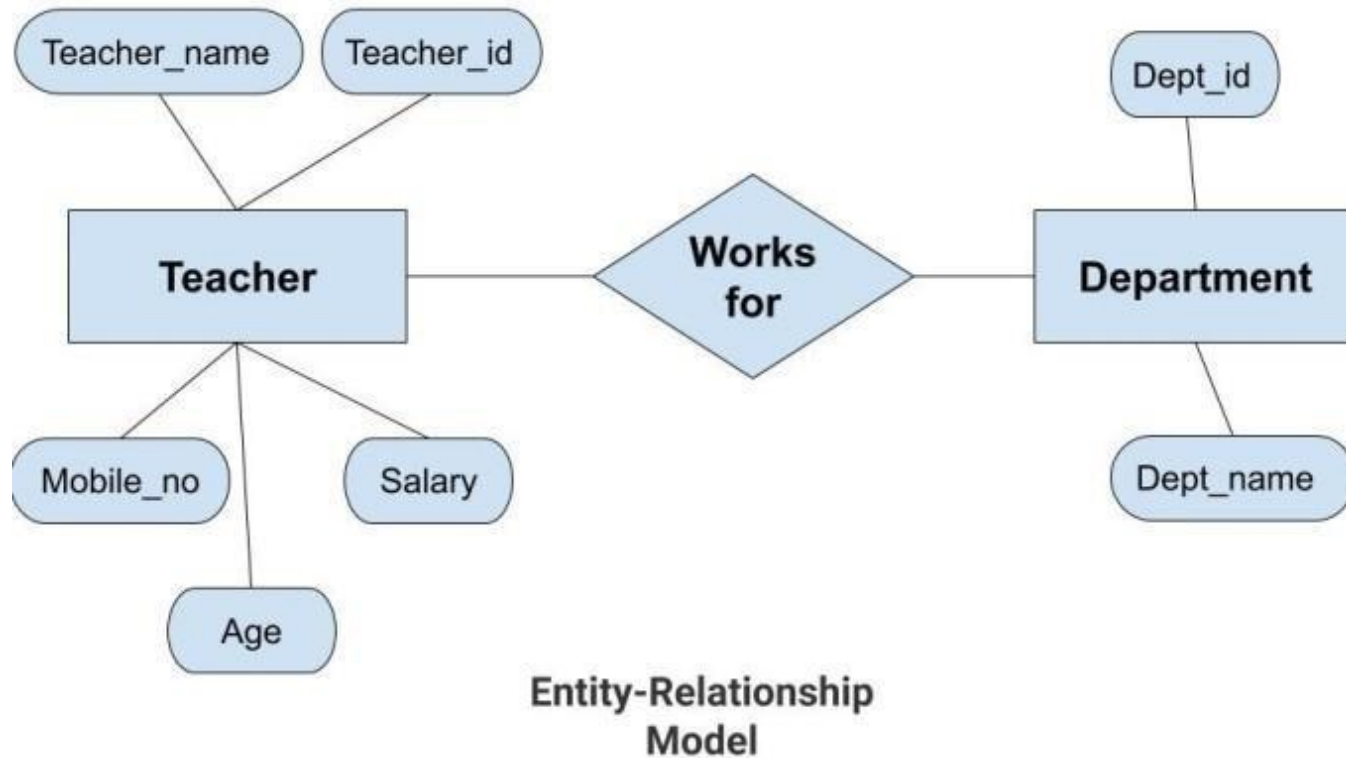


Many to Many Relationship

- When more than one instances of an entity is associated with more than one instances of another entity then it is called many to many relationship. For example, a can be assigned to many projects and a project can be assigned to many students.



Entity-relationship Model



Entity-relationship Model

- In the above diagram, the entities are Teacher and Department.
- The attributes of Teacher entity are Teacher_Name, Teacher_id, Age, Salary, Mobile_Number.
- The attributes of entity Department entity are Dept_id, Dept_name.
- The two entities are connected using the relationship. Here, each teacher works for a department.

Advantages of ER Model

- ER model is very well integrated with relational database model.
- Database main entity and their relationship are easily viewed and understood through ER model.

Disadvantages of ER Model

- No data manipulation language or commands are available in ER model.
- This model becomes crowded due to huge presence of entity.

Relational Model

- Relational Model is the most widely used model.
- In this model, the data is maintained in the form of a two-dimensional table. All the information is stored in the form of row and columns. The basic structure of a relational model is tables.
- So, the tables are also called relations in the relational model.

Relational Model

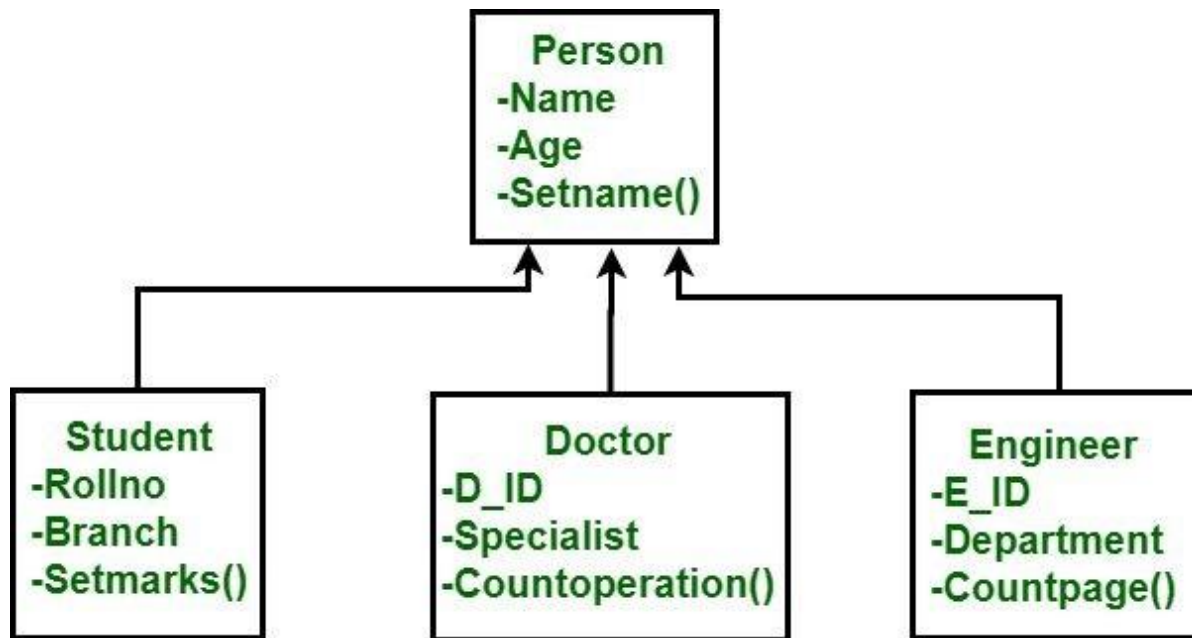
Emp_id	Emp_name	Job_name	Salary	Mobile_no	Dep_id	Project_id
AfterA001	John	Engineer	100000	9111037890	2	99
AfterA002	Adam	Analyst	50000	9587569214	3	100
AfterA003	Kande	Manager	890000	7895212355	2	65

EMPLOYEE TABLE

Object-Oriented Data Model

- The real-world problems are more closely represented through the object-oriented data model. In this model, both the data and relationship are present in a single structure known as an object.
- It is a flexible database that supports the use of abstract data types, objects and classes and that can store a wide range of data, often including sound, video and graphics.
- Codes can be reused due to inheritance. Easily understandable. Cost of maintenance can be reduced due to reusability of attributes and functions because of inheritance.

Object Oriented Data Model



SQL

- SQL stands for Structured Query Language.
- It is used for storing and managing data in relational database management system(RDMS).
- It is a standard language for Relational Database System.
- It enables a user to create, read, update and delete relational databases and tables.
- All the RDBMS like MySQL, Informix, Oracle, MS Access and SQL Server use SQL as their standard database language.

Some SQL Commands

- SELECT : extracts data from a database
- UPDATE : updates data in a database
- DELETE : deletes data from a database
- INSERT INTO : inserts new data into a database
- CREATE DATABASE : creates a new database
- CREATE TABLE : creates a new table
- DROP TABLE : deletes a table

Database Design and Data Security

- Database design provides a means to represent real world entities in a form that can be processed by the computer. It is a structural design which will be used to store and manage data.
- Database security involves protecting a database from unauthorized access, malicious destruction and even any accidental loss or misuse. Due to the high value of data incorporate databases, there is strong motivation for unauthorized users to gain access to it, for instance, competitors or dissatisfied employees.

Data Warehouse

- Data warehouse is a collection of data designed to support management decision making. It is the main repository of an organization's historical data in its corporate memory.
- The term data warehouse generally refers to the combination of many different databases across an entire enterprise.
- The main purpose of data warehouse is that a data analyst can perform complex queries and analysis, such as data mining on the information without slowing down the operational systems.

Characteristics of Data Warehouse

- Subject Oriented : Data to be organized towards the major subject areas of organization i.e. an insurance company orient data by customer, premium and claims etc.
- Integrated : Data warehouses consist of data from different variable sources integrated under one platform. This data obtained is extracted and transformed maintaining uniformity without depending on the source it was obtained from, this feature is known as Integrated.
- Non-volatile: A data warehouse is not updated in real-time. It is periodically updated via the uploading of data, protecting it from the influence of momentary change.
- Time-Variant: Data is organized via time-periods (weekly, monthly, annually, etc.

Uses/Advantages of Data Warehouse

- Easy to retrieve and find information
- More cost-effective decision making
- Better enterprise intelligence
- Fast retrieval of information
- Help management in decision making support system
- Enhance the value of Business
- Enhance customer service

Data Mining

- Generally Data mining (called knowledge discovery) is the process of analyzing data and summarizing it into useful information
- Technically, data mining is the process of finding correlations or patterns among dozens of fields in large relational databases.
- It is the principle of sorting through large amounts of data and picking out relevant information.

Advantages of Data Mining

- Automated prediction of trends and behaviours: Data mining technologies automates the process of predicting information in large databases. Statistical analytical tool can use to predict the behaviours / trends of business from analyzing large databases.
- Automated discovery of previously unknown patterns: Data mining tools can also identify hidden patterns like fraudulent credit card transaction.

Database Administrator

- The person who has control over both data and the program that accesses those data are called database administrator (DBA). The functions of database administrator are:
 - schema definition: The DBA creates the original structure of the database by using DDL.
 - schema and physical organization modification: The changes needed in any organization is analyzed and then appropriate change is made in the database schema by the DBA.

Contd.

- Functions of DBA:
 - granting of authorization: By granting different types of the authorization, DBA can regulate which part of the database various users can access.
 - runtime maintenance: DBA periodically backup the database, ensures that enough free disk space is available for normal operations.